

IP ADDRESS & SUBNETTING

List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

ANS:

Class	First Octet Range	Default Subnet Mask	Primary Purpose	Real-World Example / Use Case
Class A	1 – 126	255.0.0.0 (/8)	Massive global networks, ISPs, and large enterprises requiring millions of hosts.	Corporate Cloud Backbones / Large ISPs (e.g., Enterprise private networks using 10.0.0.0/8 or public IP ranges owned by major Telecoms).
Class B	128 – 191	255.255.0.0 (/16)	Medium to large organizations, universities, and regional ISPs.	University Campus Networks (e.g., mid-sized academic networks using 172.16.0.0/16 to assign IPs across departments and labs).
Class C	192 – 223	255.255.255.0 (/24)	Small business local networks and residential LANs.	Home Wi-Fi Routers & POS Terminals (e.g., local network interfaces assigned 192.168.1.x addresses).

Class D	224 – 239	N/A (Multicast)	One-to-many data distribution (Multicasting) rather than individual host endpoints.	IPTV Live Streams & Routing Protocols (e.g., OSPF router updates or live media distribution protocols using 224.0.0.x).
Class E	240 – 255	N/A (Reserved)	Reserved for research, testing, and experimental purposes.	Internet R&D / Experimental Testbeds (used strictly in networking research labs and non-routable environments).

Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.
Hint: Show your working steps for each calculation.

ANS:

1. Identify Network and Host Portions

- **IP Address:** 192.168.10.45
- **Subnet Mask:** 255.255.255.0 (or /24 in CIDR notation)

The subnet mask 255.255.255.0 specifies that the first 3 octets (255.255.255) represent the **Network ID**, while the last octet (0) represents the **Host ID**.

2. Network Address Calculation

To calculate the Network Address, perform a bitwise AND operation between the IP address octets and the subnet mask octets (which keeps the network portion unchanged and sets all host bits in the 4th octet to 0):

$$\begin{array}{rcl} \text{IP (4th Octet)} & : & 00101101_2 \quad (45) \\ \text{Mask (4th Octet)} & : & 00000000_2 \quad (0) \\ \hline \text{Network Octet} & : & 00000000_2 \quad (0) \end{array}$$

- **Network Address:** 192.168.10.0

3. Broadcast Address Calculation

To find the Broadcast Address, set all binary host bits in the 4th octet to 1 (which equals 255 in decimal):

$$\begin{aligned} \text{Network Octet: } & \quad 00000000_2 \quad \text{Host Bits set to 1: } \quad 11111111_2 \quad (255) \\ \text{Broadcast Address: } & \quad \mathbf{192.168.10.255} \end{aligned}$$

4. Valid Host Address Range

The usable host addresses lie strictly between the Network Address and the Broadcast Address:

- **First Valid Host:** Network Address + 1 = **192.168.10.1**
- **Last Valid Host:** Broadcast Address – 1 = **192.168.10.254**
- **Valid Host Range:** **192.168.10.1 to 192.168.10.254** (Total of 254 usable IPs)

Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

ANS:

To divide the **10.0.0.0/24** network into **4 equal subnets**, we need to borrow **2 bits** from the host portion ($2^2 = 4$ subnets).

Subnetting Mechanics

- **Original Network:** **10.0.0.0/24** (Subnet Mask: **255.255.255.0**, 256 total IP addresses)
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- **New Prefix Length:** **/26** ($24 + 2 = 26$ bits)
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- **New Subnet Mask:** **255.255.255.192**
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- **Block Size (Increment):** 64 IP addresses per subnet ($256/4 = 64$)
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- **Usable Hosts per Subnet:** $64 - 2 = 62$ hosts

Subnet Allocation Plan

Department / Group	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127

Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. **Constraint:** Show your calculation steps and final answers.

ANS:

To create **at least 20 subnets** from the base network **172.16.0.0/16**, we must borrow bits from the host portion of the address.

Step 1: Calculate the Number of Borrowed Bits

To determine how many bits (n) to borrow, use the formula $2^n \geq \text{requiredsubnets}$:

- For $n = 4$: $2^4 = 16$ subnets (not enough)
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- For $n = 5$: $2^5 = 32$ subnets (sufficient for 20 subnets)
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We must borrow **5 bits** from the host portion.

Step 2: Calculate the New Subnet Mask (CIDR Notation & Standard Form)

1. **New CIDR Notation:**
- 2.
- 3.
4. $\text{OriginalCIDR} + \text{BorrowedBits} = 16 + 5 = /21$

5. **Convert /21 to Standard Subnet Mask:**

6.

- A /21 prefix means 21 network bits followed by 11 host bits (total 32 bits):

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$$\text{○ } 11111111.11111111.11111000.00000000_2$$

- Converting to decimal octets:

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- 1st octet: $11111111_2 = 255$

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- 2nd octet: $11111111_2 = 255$

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- 3rd octet: $11111000_2 = 128 + 64 + 32 + 16 + 8 = 248$
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- 4th octet: $00000000_2 = 0$
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- New Subnet Mask: **255.255.248.0**
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Step 3: Calculate the Number of Usable Hosts per Subnet

1. Determine Remaining Host Bits (h):
- 2.
- 3.
4. $h = 32 - 21 = 11bits$
5. Apply Usable Host Formula ($2^h - 2$):
- 6.
- 7.
8. $TotalIPaddressespersubnet = 2^{11} = 2,048$
- 9.
10. $UsableHosts = 2,048 - 2 = 2,046$
11. (Subtracting 2 accounts for the reserved Network Address and Broadcast Address per subnet).
- 12.

Summary of Final Answers

- New CIDR Prefix: **/21**
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- New Subnet Mask: **255.255.248.0**
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- Total Subnets Created: **32** (meets requirement of at least 20)
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- Usable Hosts per Subnet: **2,046**

A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

ANS:

The reason your friend's device isn't connecting is that **192.168.1.300 is an invalid IP address.**

The Error

An IPv4 address is made up of four octets (numbers separated by dots), and each octet is represented by 8 bits in binary. This means the value for any single octet can only range from **0 to 255**. The number **300** exceeds this maximum limit, so no network device will recognize or assign it.

What a Valid IP Address Looks Like

In a standard home network using the **192.168.1.0/24** subnet (subnet mask **255.255.255.0**):

- **Valid Host Range:** Any number from **192.168.1.1** through **192.168.1.254**

- **Example Fix:** Changing the address to something like **192.168.1.30** or setting the device to **DHCP** (so the router automatically assigns a valid IP) will resolve the connection issue immediately.